Mini Assessment

Vector Databases and Semantic Search

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Q1. What is the purpose of a vector database?

Answer: A vector database stores numerical vector representations of items such as text, images, or audio. It supports similarity search, allowing an application to retrieve items whose vectors are close to a query vector. This is useful for semantic search, recommendation systems, and retrieval-augmented generation.

Q2. Why are embeddings useful for semantic search?

Answer: Embeddings represent the meaning or features of content as numerical vectors. Texts with related meanings may have vectors that are close to each other even when they do not use exactly the same words. A search system can compare these vectors to find conceptually relevant results.

Q3. What does cosine similarity measure?

Answer: Cosine similarity measures the angle-based similarity between two non-zero vectors. A value closer to 1 generally indicates that the vectors point in similar directions, a value near 0 indicates little directional similarity, and a value closer to -1 indicates opposite directions. Its interpretation depends on the embedding model and data.